Mingyuan Zhang

Boston, MA Homepage zhang.mingyua@northeastern.edu 781 219 1785 Google Scholar

EDUCATION

***Northeastern University* (NEU)** Boston, MA
• **Ph.D.** candidate in **Computer Engineering** Sep. 2023 – May 2028 (expected)
• Advisor: Prof. Yun Raymond Fu (Member of the Academy of Europe; Fellow of NAI, AAAS, AAAI, IEEE, AIMBE, OSA, SPIE, IAPR)
• GPA: 3.7/4.0 • Research Assistant at SmileLab • Focusing on computer vision, NLP and machine learning

***Northeastern University* (NEU)** Boston, MA
• **Master of Science** in **Electrical and Computer Engineering** Sep. 2021 – Dec. 2022
• Research Assistant at SmileLab
• **Concentration:** Computer Vision, Machine Learning and Algorithm

***Xidian University* (XDU)** Xi'an, China
• **Bachelor of Engineering** in **Telecommunications Engineering** Sep. 2017 – Jun. 2021
• Honor of Excellent Students Leader, 2020

RESEARCH INTEREST

Computer Vision and Artificial Intelligence | Performance, Efficiency, and Generalizability in Foundation Models | Image/Video Understanding | Motion/Trajectory Prediction and Planning | AI Agents | Multimodal Learning | Image/Video Generation

EXPERIENCE

Northeastern University | **SmileLab** | *Research Assistant* | Boston, MA *Sep. 2023 - Present*

Advisor: Prof. Yun Raymond Fu

- **Large Language Models and Multimodal Learning**
  - Proposed Mask Fine-Tuning (MFT), a novel *parameter-efficient adaptation paradigm* that improves large language and vision-language models, consistently outperforming even full fine-tuning.
  - Developed a structured *latent Mixture-of-Experts (MoE)* framework for large language models to *reduce inference FLOPs* while maintaining or even improving model performance.
  - Conducted SoupLM, a learnable model integration framework that *merges multiple large language and multimodal models* into a unified model with improved generalization and negligible inference overhead.
  - Co-developed a plug-and-play keyframe selection agent for video VLMs that uses CLIP/SigLIP-based semantic tools to provide informative visual priors for improved reasoning.
  - Contributed to Cautious Next Token Prediction (CNTP), an entropy-aware decoding strategy that *improves reasoning accuracy and reliability of large language models.*
  - Collaborated on a systematic survey on *hallucination in video large language models*, identifying root causes and advancing research toward more reliable multimodal systems.
- **Pruning and Distillation**
  - Proposed Sharpness-Aware Random Sparse Training (SA-RST) and Dual Lottery Ticket Hypothesis, enabling *effective training of randomly initialized sparse neural networks* without dense pre-training.
  - Leading a comprehensive survey on *model/data pruning and distillation aimed at improving model performance* for foundation models.
- **Autonomous Driving and Robotics**
  - Developing a backend-agnostic *visual evidence agent using RL* for autonomous driving to *improve VLA decision quality* under strict latency and token budgets.
  - A survey investigating the integration of *large language models into trajectory prediction*, identifying key challenges and advancing language-driven motion understanding for autonomous systems.
- **Generative Models**
  - Contributed to an *agent-centric framework for visual generation* that leverages explicit structural guidance to bridge high-level intent and low-level synthesis.
  - Developed LST-GAN, a *spatial-temporal generative framework that synthesizes missing camera views* to improve robustness and accuracy in 3D human pose estimation.

Socure | **Computer Vision team** | *Research Intern* | Remote *Jun. 2025 – Aug. 2025*

Mentor: Dr. Weqiang Mu, Dr. Feng Xiao

- **Low-Level Vision**
  - Developed a computational framework for barcode reconstruction aimed at mitigating the effects of occlusion, blur, and partial data loss, thereby enhancing decoding robustness and accuracy.

AinnovationLabs | *Applied Scientist Intern* | Remote *Dec. 2022 – Dec. 2024*

Mentor: Prof. Yun Raymond Fu

- **Healthcare AI and App Development**
  - Invented **1 patent** for sleep signal processing that identifies sleep stages and detects sleep disorders from complex acoustic signals.
  - Developed a mobile app for recording sleep patterns and consulting sleep-related questions, enabling large-scale multimodal data collection for sleep research.
- **Computer Vision and Human-Centered Perception**
  - Developed a spatio-temporal 3D human pose estimation model with a self-collected dataset for accurate single-person motion tracking.
  - Built a pedestrian re-identification system capable of recognizing the same individual across diverse environments.

Southeast University | *Research Assistant* | Nanjing, China *Jun. 2019 - Aug. 2019*

Advisor: Prof. Fengqi Si

- **Environmental Modeling**
  - Designed a prediction model for atmospheric environmental changes by integrating multimodal environmental data, improving forecasting accuracy and robustness.

PUBLICATION

Under Review / Preprint

[U-1] ***Mingyuan Zhang***, Yue Bai, Huan Wang, Yizhou Wang, Qihua Dong, Yitian Zhang, and Yun Fu, "Boosting Large Language Models with Mask Fine-Tuning". [Paper] [Code]

[U-2] ***Mingyuan Zhang***, Yue Bai, Yifan Wang, Yiyang Huang, and Yun Fu, "Rethinking Fine-Tuning: Unlocking Hidden Capabilities in Vision-Language Models". [Paper]

[U-3] Yi Xu, Ruining Yang, Yitian Zhang, Yizhou Wang, Jianglin Lu, ***Mingyuan Zhang***, Lili Su, and Yun Fu, "Trajectory Prediction Meets Large Language Models: A Survey". [Paper] [Code]

[U-4] Yiyang Huang, Yitian Zhang, Yizhou Wang, ***Mingyuan Zhang***, Liang Shi, Huimin Zeng, and Yun Fu, "Distorted or Fabricated? A Survey on Hallucination in Video LLMs". [Paper] [Code]

[U-5] Yue Bai*, ***Mingyuan Zhang****, Huan Wang, Zhiqiang Tao, Kunpeng Li, and Yun Fu, "Random Sparse Networks Training with Sharpness-Aware Regularization".

[U-6] Yue Bai*, ***Mingyuan Zhang****, Handong Zhao, and Yun Fu, "Advancing Multi-Modal Foundation Models with Adapter Ensemble Strategies".

Conference

[C-1] Yizhou Wang, Lingzhi Zhang, Yue Bai, Mang Tik Chiu, Zhengmian Hu, ***Mingyuan Zhang***, Qihua Dong, Yu Yin, Sohrab Amirghodsi, and Yun Fu, "Cautious Next Token Prediction", ***ACL 2025***. [Paper] [Code]

Ongoing Work

[O-1] ***Mingyuan Zhang***, Yue Bai, Yi Xu, Chang Liu, and Yun Fu, "LST-GAN: Latent Spatio-Temporal Generative Adversarial Networks for Missing-View 3D Human Pose Estimation".

[O-2] Xu Ma, Jianglin Lu, ***Mingyuan Zhang***, and Yun Fu, "Agentic Visual Generation in the Late 2020s".

[O-3] Yue Bai*, ***Mingyuan Zhang****, Jiasen Lu, Zichen Zhang, and Yun Fu, "SoupLM: Model Integration in Large Language and Multi-Modal Models".

HONOR & AWARD

- 5th Internet+ Innovation and Entrepreneurship Competition **Provincial Bronze Medal**

ACADEMIC SERVICE

Reviewer:

- Conference: ACMMM 2024, AAAI 2025, UAI 2025, ACMMM 2025, NeurIPS 2025
- Journal: TKDD, TPAMI

Teaching:

- TA of EECE5639 Computer Vision, Spring 2026, NEU BOS

SKILL

Languages: Strong reading, writing and speaking competencies for English, Mandarin Chinese.

Coding: Python, MATLAB, C, Xcode.

Web Dev: HTML, CSS.

Misc.: Academic research, LaTeX typesetting and publishing.